Chapter 1: Combinatorial Analysis

1.1 Introduction

Combinatorial analysis is the mathematical theory of counting. It provides effective methods for counting the number of ways things can occur, which is fundamental to probability theory.

- **Core Concept:** Many probability problems can be solved by counting the number of outcomes in a specific event and dividing it by the total number of possible outcomes.
- **Example (Antenna System):**
    - **Problem:** n antennas are lined up. m are defective. The system works if no two consecutive antennas are defective.
    - **Method:** Count the functional configurations and divide by the total configurations to find the probability.

1.2 The Basic Principle of Counting

This is the fundamental rule underlying most combinatorial work.

The Basic Principle (Two Experiments)

If Experiment 1 has m possible outcomes and, for *each* of those outcomes, Experiment 2 has n possible outcomes, then the total number of combined outcomes is:

$$\text{Total Outcomes} = m \times n$$

- **Proof:** The set of outcomes is a matrix of m rows and n elements, enumerated as:
  $(1,1), (1,2), \ldots, (1,n)$
  $\ldots$
  $(m,1), (m,2), \ldots, (m,n)$

The Generalized Basic Principle

If there are r experiments performed sequentially:

1. Experiment 1 has n_1 outcomes.
2. Experiment 2 has n_2 outcomes for each outcome of Experiment 1.
3. ...
4. Experiment r has n_r outcomes for each previous combination.

Then the total number of outcomes is:

$$\text{Total Outcomes} = n_1 \cdot n_2 \cdot n_3 \cdots n_r$$

- **Example (License Plates):** A 7-place plate with 3 letters followed by 4 numbers.
    - Calculation: $26 \cdot 26 \cdot 26 \cdot 10 \cdot 10 \cdot 10 \cdot 10 = 175,760,000$.
    - *If no repetitions allowed:* $26 \cdot 25 \cdot 24 \cdot 10 \cdot 9 \cdot 8 \cdot 7$.

1.3 Permutations

A **permutation** is an ordered arrangement of objects.

Permutations of Distinct Objects

For n distinct objects, the number of different ordered arrangements is $n!$ (n-factorial).

$$n! = n(n-1)(n-2)\cdots 3 \cdot 2 \cdot 1$$

- **Reasoning:** You have n choices for the 1st spot, $(n-1)$ for the 2nd, down to 1 choice for the last spot.
- **Example:** Batting orders for 9 players = $9! = 362,880$.

Permutations with Indistinguishable Objects

When a set of n objects contains items that are identical to one another, we must divide out the "duplicate" arrangements.

If there are n objects where:

- n_1 are alike (type 1)
- n_2 are alike (type 2)
- ...
- n_r are alike (type r)

The number of different permutations is:

$$\frac{n!}{n_1!n_2!\cdots n_r!}$$

- **Example (PEPPER):**
  - Total letters $n = 6$.
  - P's ($n_1 = 3$), E's ($n_2 = 2$), R ($n_3 = 1$).
  - Arrangements = $\frac{6!}{3!2!1!} = 60$.

1.4 Combinations

A **combination** represents a group of objects where the **order of selection is irrelevant**.

Formula

The number of different groups of size r selected from n distinct objects is denoted as $\binom{n}{r}$ (read as "n choose r").

$$\binom{n}{r} = \frac{n!}{(n-r)!r!}$$

- **Logic:** There are $n(n-1)\cdots(n-r+1)$ ways to select the objects *in order*. Since the order doesn't matter for a group, we divide by $r!$ (the number of ways to arrange the group itself).
- **Definition:** By convention, $0! = 1$, so $\binom{n}{0} = \binom{n}{n} = 1$.

Key Identities and Theorems

1. **Combinatorial Identity:**

$$\binom{n}{r} = \binom{n-1}{r-1} + \binom{n-1}{r}$$

   - *Interpretation:* A specific object is either *in* the group (leaving $r-1$ spots for the remaining $n-1$ items) or *not in* the group (leaving r spots for the remaining $n-1$ items).

2. **The Binomial Theorem:**

$$(x+y)^n = \sum_{k=0}^{n} \binom{n}{k} x^k y^{n-k}$$

   - The coefficients $\binom{n}{k}$ are called **binomial coefficients**.
   - *Example:* $(x+y)^3 = y^3 + 3xy^2 + 3x^2y + x^3$.

3. **Total Number of Subsets:**
   The total number of subsets of a set with n elements is 2^n.

$$\sum_{k=0}^{n} \binom{n}{k} = 2^n$$

1.5 Multinomial Coefficients

This generalizes combinations to dividing items into more than two groups.

Division into Groups

If n distinct items are to be divided into r distinct groups of sizes $n_1, n_2, \ldots, n_r$ (where $\sum n_i = n$), the number of possible divisions is:

$$\binom{n}{n_1, n_2, \ldots, n_r} = \frac{n!}{n_1! n_2! \cdots n_r!}$$

This quantity is called a **multinomial coefficient**.

- **Note:** This is mathematically identical to the formula for permutations of indistinguishable objects.
- **Example:** A police department of 10 officers splitting into groups of 5 (patrol), 2 (station), and 3 (reserve).
  - Calculation: $\frac{10!}{5!2!3!} = 2520$.

The Multinomial Theorem

$$(x_1 + x_2 + \cdots + x_r)^n = \sum_{(n_1, \ldots, n_r): \sum n_i = n} \binom{n}{n_1, \ldots, n_r} x_1^{n_1} x_2^{n_2} \cdots x_r^{n_r}$$

The sum is taken over all non-negative integer vectors $(n_1, \ldots, n_r)$ that sum to n.

1.6 The Number of Integer Solutions of Equations

This section deals with distributing n **indistinguishable** items into r **distinguishable** bins (or urns). This is equivalent to finding integer solutions to sums.

Proposition 6.1 (Positive Solutions)

The number of distinct **positive** integer-valued vectors $(x_1, \ldots, x_r)$ satisfying $x_1 + \cdots + x_r = n$ (where $x_i > 0$) is:

$$\binom{n-1}{r-1}$$

- **Visual Proof:** Imagine n objects in a line 00000. There are $n-1$ spaces between them. To create r groups, you choose $r-1$ spaces to place dividers.

Proposition 6.2 (Non-negative Solutions)

The number of distinct **non-negative** integer-valued vectors $(x_1, \ldots, x_r)$ satisfying $x_1 + \cdots + x_r = n$ (where $x_i \geq 0$) is:

$$\binom{n+r-1}{r-1}$$

- **Derivation:** Let $y_i = x_i + 1$. Then $y_i > 0$. The equation becomes $\sum y_i = n + r$. Using Prop 6.1, the answer is $\binom{(n+r)-1}{r-1}$.
- **Example (Investment):**
  - Invest \$20k in 4 investments (x_1, x_2, x_3, x_4).
  - Equation: $x_1 + x_2 + x_3 + x_4 = 20$ (where $x_i \geq 0$).
  - Calculation: $\binom{20+4-1}{4-1} = \binom{23}{3} = 1771$ strategies.

Application: Non-Consecutive Defectives

Revisiting the Example from Section 1.1 using this method:

- n items total, m defective, $n-m$ functional.
- How many orderings where no two defectives are consecutive?
- Formula derived: $\binom{n-m+1}{m}$.